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RESOLUTION MSC.581(110) (adopted on 27 June 2025)
REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS

ANNEX 28
RESOLUTION MSC.581(110)
(adopted on 27 June 2025)
REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee
RECALLING ALSO its adoption, by resolution A.864(20), of the Recommendations for entering enclosed spaces aboard ships , and by resolution A.1050(27), of the Revised recommendations for entering enclosed spaces aboard ships , incorporating therein recommendations for entering cargo spaces, tanks, pump-rooms, fuel tanks, cofferdams, duct keels, ballast tanks and similar enclosed spaces,
RECALLING FURTHER that SOLAS regulation III/19.3.6 requires enclosed space entry and
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keels, ballast tanks and similar enclosed spaces,
RECALLING FURTHER that SOLAS regulation III/19.3.6 requires enclosed space entry and rescue drills, SOLAS regulation XI-1/7 requires the carrying of atmosphere testing instrument for enclosed spaces, and SOLAS chapter IX and the ISM Code require management for the safe operation of ships,
RECALLING that the Maritime Safety Committee is requested to keep the revised recommendations under review and amend them as necessary,
BEING CONCERNED about the continued loss of life resulting from personnel entering shipboard spaces in which the atmosphere is unable to support life, owing to oxygen depletion, oxygen enrichment, toxicity or flammability,
BEING AWARE of the work undertaken in this regard by the International Labour Organization, Governments and segments of the private sector,
RECOGNIZING that organizational leadership plays a crucial role in successful
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Governments and segments of the private sector,
RECOGNIZING that organizational leadership plays a crucial role in successful implementation of this guidance by empowering shipboard staff to make the right decisions,
HAVING CONSIDERED , at its 110th session, the recommendation made by the Sub-Committee on Carriage of Cargoes and Containers at its tenth session,
1 ADOPTS the Revised recommendations for entering enclosed spaces aboard ships, as set out in the annex to the present resolution;
2 INVITES Governments to bring the annexed Revised Recommendations to the attention of shipowners, ship operators, seafarers, port and terminal operators and port workers, urging them to apply them, as appropriate, to all ships and operations;
3 INVITES the Assembly to revoke resolution A.1050(27) and to endorse the action taken by the Maritime Safety Committee.

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ANNEX
REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS
Preamble
The objective of these recommendations is to encourage the adoption of safety procedures and the development of sound practices aimed at preventing casualties to, and enhancing the safety of, personnel entering or working in enclosed spaces where there may be an oxygen-deficient, oxygen-enriched, flammable and/or toxic atmosphere. Additionally, in the event of an emergency within an enclosed space, these recommendations provide guidance for an appropriate, planned and considered response complying with ship-specific enclosed space rescue plans.
Investigations into the circumstances of enclosed space accidents have shown that a failure to
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Investigations into the circumstances of enclosed space accidents have shown that a failure to systematically identify the hazards, assess the risks and implement an appropriate entry procedure remains a significant factor in many accidents. Likewise, the complex structure of some spaces on board ships creates problems with appropriate ventilation, illumination and movement within the space. Organizational leadership on ship and shore plays a crucial role in successful implementation of this guidance by empowering shipboard staff to make the right decisions. These recommendations apply to all types of ships and provide guidance to ship operators, seafarers and shore personnel to work on board ships. It should be noted that on ships where entry into enclosed spaces may be infrequent, for example, on certain pas senger ships or small general cargo ships, the dangers may be less apparent, and accordingly there may be a need for increased vigilance.

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small general cargo ships, the dangers may be less apparent, and accordingly there may be a need for increased vigilance.

These recommendations are intended to complement national laws or regulations, accepted standards or particular procedures which may exist for specific trades, ships or types of shipping operations.

It may not be practicable to apply all of these recommendations to all situations; however, when applying these recommendations becomes impracticable, every endeavour should be made to observe the intent of the recommendations, and attention should be paid to the hazards that may be involved in the specific entry and mitigations required to reduce the risks to an acceptable level.

1 INTRODUCTION

The atmosphere in any enclosed space may be oxygen -deficient or oxygen-enriched and/or contain flammable and/or toxic gases or vapours. Such unsafe atmospheres could also subsequently occur in a space previously found to be safe. Unsafe atmospheres are lik ely to

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subsequently occur in a space previously found to be safe. Unsafe atmospheres are lik ely to be present in spaces that are connected to a space containing a hazardous atmosphere and may also be present in spaces adjacent to those spaces where a hazard is known or suspected to be present.

1.1 An unsafe atmosphere can also quickly occur in a space previously found to be safe, including cargo holds and tanks where the hatch covers and tank covers have been open for some time. Unsafe atmospheres can occur in frequently visited working areas and stores, such as forecastle head spaces, pump-rooms, compressor rooms, inert gas rooms and other spaces that are adjacent to, or connected to, spaces containing a hazardous atmosphere.

These spaces include cargo holds containing cargo that depletes oxygen and /or emits toxic, flammable or explosive gases, or is under fumigation.

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flammable or explosive gases, or is under fumigation.

1.2 An unsafe atmosphere can also occur in void spaces, cofferdams, pipe tunnels and also in accommodation areas and engine -rooms adjacent to or connected to cargo holds containing hazardous cargo or under fumigation due to a failure in the means of sealing the connection between the cargo space and the adjacent spaces.
1.3 The emission of toxic, flammable or explosive gases from cargo tanks, fuel tanks, slop tanks and sewage tanks can also give rise to a potentially dangerous situation around open access hatches, manholes and ventilators on deck.
2 DEFINITIONS
2.1 Enclosed space means a space which may contain a hazardous atmosphere or lack of oxygen and has any of the following characteristics:

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2.1 Enclosed space means a space which may contain a hazardous atmosphere or lack of oxygen and has any of the following characteristics:

.1 limited openings for entry and exit;

.2 inadequate ventilation; or

.3 not designed for continuous human occupancy.

This includes but is not limited to spaces that are diverse in their characteristics such as cargo holds, bilge spaces, ballast and other tanks, pump -rooms, chain lockers and engine crankcases.

2.2 Connected space means a space that is connected, by either permanent or temporary means (such as a door), to a source space that may contain a hazardous atmosphere. For clarity, a space separated by a manual door, even if watertight, should be considered as "connected" as it is impossible to tell from outside the space whether it is open or closed or indeed properly sealed. A connected space should be treated as containing a hazardous atmosphere until testing proves otherwise. The nature of the connection may lead to a

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atmosphere until testing proves otherwise. The nature of the connection may lead to a trapped hazardous atmosphere.

2.3 Adjacent space means a space that shares a common boundary with a compartment that may contain a hazardous atmosphere. Such a space has no openings, temporary or permanent, into the hazardous compartment whatsoever and is designed to be a contiguous barrier. Such a space may only contain a hazardous atmosphere in the event of failure of that barrier. Precautions should relate to the possibility of such a failure.

2.4 Trapped Hazardous Atmosphere means a hazardous atmosphere that may be trapped in a connected space in a manner that causes that spaces atmosphere to fill and/or to empty at a different rate to the source space. Such a space, while recognized as containing the same atmosphere, should be treated independently to the source space, and should be assumed to contain a hazardous atmosphere until proved otherwise by testing. For example,

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assumed to contain a hazardous atmosphere until proved otherwise by testing. For example, a trapped atmosphere may remain even after the cargo in the source space is discharged.

2.5 Competent person means a person with an operational level of competency to make an informed assessment of the likelihood of a dangerous atmosphere being present or subsequently arising in the space.

2.6 Responsible person means a person in a management level on board a ship (i.e. master, chief mate, chief engineer officer or second engineer officer) of competency and authorized by the shipping company to permit entry into an enclosed space.

2.7 Attendant means a person maintains a watch over those entering the enclosed space, to maintain communications with those inside the space and to initiate the emergency

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to maintain communications with those inside the space and to initiate the emergency procedures in the event of an incident occurring.

2.8 Enclosed Space Register means a ship -specific register which lists all enclosed spaces on board the ship, along with their connected spaces and adjacent spaces, their hazards, associated risk mitigations if applicable, and how the atmosphere in these spaces may change depending upon the nature of cargo carried or the content of the space, and which forms a part of the safety management for enclosed spaces.

3 SAFETY MANAGEMENT FOR ENTRY INTO ENCLOSED SPACES

3.1 A safety strategy should be adopted in order to prevent accidents on entry into enclosed spaces in a comprehensive manner by the company, in consultation with the ship. This should include an Enclosed Space Register, which should be produced on a ship-by-ship basis to identify enclosed spaces, the hazards of those spaces, assessment of risks under the

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basis to identify enclosed spaces, the hazards of those spaces, assessment of risks under the differing conditions likely to arise in the space and the risk mitigation measures required to be put into place prior to entry. This should also include an assessment of how the atmosphere in these enclosed spaces may be impacted by the contents of the spaces themselves, such as in case a ship stores treated sewage or grey water temporarily in its ballast water tanks, or the contents of connected spaces or adjacent spaces, including cargo, fumigants, fuel oils, slops, oxygen-depleting conditions and the physical or structural arrangement of the space.

3.2 The company should ensure that all relevant information relating to the hazards of the cargo, as submitted by the shipper in accordance with the applicable requirements of SOLAS regulation VI/2, the International Maritime Solid Bulk Cargoes Code (IMSBC Code), the

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regulation VI/2, the International Maritime Solid Bulk Cargoes Code (IMSBC Code), the International Maritime Dangerous Goods Code (IMDG Code), the IBC Code and the International Gas Carrier Code (IGC Code), is provided in a format that is understandable to the ships crew and distributed to those on board who may be exposed to these hazards.

3.3 The company should ensure that the procedures for entering enclosed spaces are included and implemented among the key shipboard operations concerning the safety of the personnel and the ship, in accordance with paragraphs 6.4, 7 and 12.3 of the International Safety Management (ISM) Code.

3.4 The company should ensure that all relevant crew members are given adequate training in the safety management of enclosed spaces as per paragraphs 6.3 and 6.5 of the ISM Code.

3.5 The company should ensure that adequate time has been allowed for any planned

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ISM Code.
3.5 The company should ensure that adequate time has been allowed for any planned enclosed space activity, and that undue time pressure, either explicit or implied, is avoided as this has been found to be a causal factor of many enclosed space accidents.
3.6 The company should elaborate a procedural implementation scheme which provides for comprehensive training in the maintenance, calibration and use of atmospheric testing equipment in such spaces. This training should be recorded, and individual crew member s who have been deemed competent in the maintenance, calibration and use of atmospheric testing equipment should be listed.
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3.6.1 The competent and the responsible persons should be trained in enclosed space
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3.6.1 The competent and the responsible persons should be trained in enclosed space hazard recognition, evaluation, measurement, control and elimination, using standards acceptable to the Administration. The competent person should have received adequate training along with sufficient theoretical knowledge and practical experience to make an informed assessment of the likelihood of a dangerous atmosphere being present or subsequently arising in the space. The responsible person should have sufficient knowle dge of the procedures to be established and complied with on board and received adequate training, in order to ensure that an enclosed space is safe for entry and occupancy. The attendant should be adequately trained within the safety management system.
3.6.2 All crew members should be trained, as appropriate, in enclosed space safety,
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3.6.2 All crew members should be trained, as appropriate, in enclosed space safety, including familiarization with onboard procedures for recognizing, evaluating and controlling hazards associated with entry into enclosed spaces.

3.7 The company should identify and provide the necessary equipment such as ventilation devices, atmosphere testing equipment, breathing apparatus and personnel recovery apparatus to facilitate safe entry and rescue from an enclosed space. The equipment provided should be suitable for the intended use. A careful study of the operational profile should be carried out and all necessary equipment should always be on board in operational condition. Consideration should be given to the appropriate use of technol ogy to assist in the hazard identification and mitigation of enclosed space entry. Crew members should be trained in the use of equipment necessary to facilitate rescue from an enclosed space and a record of this training should be maintained.

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in the use of equipment necessary to facilitate rescue from an enclosed space and a record of this training should be maintained.

3.8 The company should ensure that rescue drills from enclosed space identified in their Enclosed Space Register are undertaken regularly as required by SOLAS regulation III/19.3.6 using the equipment provided to facilitate a rescue from an enclosed space. Such drills should focus on different aspects of operations involving enclosed spaces . SOLAS regulation III/19.3.6 requires:

.1 checking and use of personal protective equipment required for entry;

.2 checking and use of communication equipment and procedures;

.3 checking and use of instruments for measuring the atmosphere in enclosed spaces;

.4 checking and use of rescue equipment and procedures; and

.5 instructions in first aid and resuscitation techniques.

3.9 Internal audits by the company and external audits by the Administration of the ship's

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3.9 Internal audits by the company and external audits by the Administration of the ship's safety management system should verify that the established procedures 1 are followed and are consistent with the safety strategy referred to in these recommendations.
3.10 The company should establish criteria to manage additional risks involved during simultaneous operations (SIMOPS) where one of the operations includes an enclosed space entry. The criteria should consider the assessment of personnel and resources availabl e in case of an emergency following an enclosed space entry.
1 Established procedures include resolution A.1050(27) as amended, the IMSBC Code, the BLU Code, the IMDG Code, the IBC Code, the IGC Code, Recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo holds (MSC.1/Circ.1264), Guidelines on tank entry for tankers using nitrogen as an inerting medium (MSC.1/Circ.1401) and other accepted standards and industry codes of
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nitrogen as an inerting medium (MSC.1/Circ.1401) and other accepted standards and industry codes of practice, as relevant.

3.11 The company should ensure that single person entry into an enclosed space is not permitted.

4 IDENTIFICATION OF THE HAZARDS AND ASSESSMENT OF RISK

4.1 The company should ensure that a risk assessment is conducted to identify all enclosed spaces on board the ship and that the identified enclosed spaces are recorded in the Enclosed Space Register , which should be maintained on board the ship as well as ashore. This Enclosed Space Register and risk assessment should be kept up to date as appropriate to ensure its continued validity, particularly after loading and during the carriage of cargoes

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to ensure its continued validity, particularly after loading and during the carriage of cargoes which may a dversely affect the safety of the atmosphere within a space. A reassessment should also be made when the contents of the space change, such as in case a ship stores treated sewage or grey water temporarily in its ballast water tanks. The company should consider the use of appropriate technology to assist in the hazard identification and mitigation. The Enclosed Space Register and risk assessment should form the basis of the development of the enclosed space emergency response plan (app endix 1). It is recommended that the enclosed space emergency response plan be reviewed after each drill so that its effectiveness can be assessed and, if necessary, improvements made.

4.2 As entries for enclosed spaces other than cargo spaces may be different, there is a need to record information for the benefit of personnel involved. Every ship should have an

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need to record information for the benefit of personnel involved. Every ship should have an Enclosed Space Register, the information within which may form the basis of a r isk assessment. The Enclosed Space Register should contain:

.1 physical layout of the space and access and egress points, including of connected spaces, if any;

.2 physical hazards in the space, e.g. vertical ladders, unguarded openings, poor lighting, wet or slippery conditions, excessive heat;

.3 connection to adjacent spaces;

.4 specific hazards within the space, for example, the effect of ballast water treatment method on the atmosphere within ballast tanks;

.5 if used, information related to additional technology, helping to determine enclosed space condition;

.6 information related to fixed and portable ventilation systems including equipment and where the equipment is stored;

.7 estimated time taken to achieve the air changes for safe entry, using forced or natural ventilation;

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.7 estimated time taken to achieve the air changes for safe entry, using forced or natural ventilation;
.8 lighting and means for temporary lighting including intrinsically safe lighting where appropriate;
.9 means for atmosphere testing;
.10 any pertinent information that would assist the risk assessment process;
.11 locking and "Safe to enter"/"Unsafe to enter" signage arrangements; and
.12 the equipment necessary to facilitate emergency rescue from the space.

4.3 In order to ensure safety from the outset, entry and occupancy, a competent person should make an assessment of any potential hazards in the space to be entered, taking into account the characteristics of the previous and current cargo carried with Safety Data Sheets

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account the characteristics of the previous and current cargo carried with Safety Data Sheets (SDS) and cargo information, ventilation of the space, coating of the space and other relevant factors. Opening of an enclosed space should be subject to a risk assessment, taking into consideration the potential for release of dangerous ga ses or potential to create an explosive atmosphere: the competent person's assessment should determine the potential for the presence of an oxygen -deficient, oxygen -enriched, flammable or toxic atmosphere, which includes carbon monoxide (CO) and carbon dioxide (CO2) as well as other toxic or asphyxiant gases. The competent person should bear in mind that the ventilation procedures for an adjacent or connected space may be different from those for the enclosed space itself. The details of the assessment should be recorded in a standard format, and maintained on board the ship.

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details of the assessment should be recorded in a standard format, and maintained on board the ship.

4.4 The nature of the hazards that may be present should be fully understood by those on board the ship, both crew and shore-based personnel, and those ashore managing the ship.

4.5 The procedures to be followed for testing the atmosphere in the space and for entry should be decided on the basis of the assessment. These will depend on whether the assessment shows that:

.1 there is minimal risk to the health or life of personnel entering the space; or

.2 there is no immediate risk to health or life, but a risk could arise during the course of work in the space; or

.3 a risk to health or life is identified.

4.6 Where the assessment indicates minimal risk to health or life or potential for a risk to arise during the course of work in the space, the precautions described in sections 5 to 8 and 10 should be followed, as appropriate.

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arise during the course of work in the space, the precautions described in sections 5 to 8 and 10 should be followed, as appropriate.

4.7 Where the assessment identifies a risk to life or health, if entry is to be made, the additional precautions specified in section 9 should also be followed.

4.8 Throughout the assessment process, there should be an assumption that the space to be entered is considered as hazardous until positively proved to be safe for entry.

5 AUTHORIZATION OF ENTRY

5.1 No person should open or enter an enclosed space unless authorized by the master or a responsible person nominated by the master and the appropriate safety procedures laid down for the particular ship have been followed.

5.2 Entry into enclosed spaces should be planned and the use of an entry permit system, which may include the use of a checklist, should always be used where the assessment

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which may include the use of a checklist, should always be used where the assessment identifies any risk to health or life. Prior to the entry of a space, an enclosed space entry permit should be issued by the master or the responsible person nominated by the master, and complied with at all times by the competent person included in the work especially those who enter the space. An example of the enclosed space entry permit is provided in appendix 2. The validity of the permit should be specified based on the risk assessment and should never be longer than eight hours.

5.3 Additional permits may be required depending upon the ships safety management system (SMS) and the activity being planned within the enclosed space. These permits may
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system (SMS) and the activity being planned within the enclosed space. These permits may include energy isolation permits, electrical isolation permit, hot work permit, working at height permit or the completion of documents such as Job Hazard Analysis.

5.4 Shore personnel should be provided with information on the hazards associated with enclosed spaces that may be present aboard the ship.

6 GENERAL PRECAUTIONS

6.1 Access to enclosed spaces should be carefully managed on ships where shore personnel regularly enter cargo spaces to load and discharge cargoes, especially on bulk carriers and general cargo ships. These recommendations are complementary to the requirements of the International Ship and Port Facility Security (ISPS) Code. It is recommended that a basic schematic plan of the ships spaces should be placed at the gangway or at other access points of the ship clearly indicating which spaces are safe to enter,

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gangway or at other access points of the ship clearly indicating which spaces are safe to enter, and which spaces are not. The plan should outline the enclosed space entry procedures that should be complied with by both ship and shore personnel before any entry is permitted. The use of simple safe to enter and unsafe to enter symbols is recommended. The plan should be time-stamped and dated and amended as necessary by the responsible person. This may also be beneficial on ships where large numbers of shore personnel are bei ng employed on the ship, such as in dry dock or ship repair facilities. An example of a ships schematic plan is contained in appendix 3.

6.2 Before any personnel are authorized to enter any cargo space containing any cargo, the competent person should carry out a risk assessment of the cargo; identification of the physical characteristics of the cargo space(s) concerned; and the operations to be carried out, and the responsible person should:

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physical characteristics of the cargo space(s) concerned; and the operations to be carried out, and the responsible person should:

.1 where it is necessary for shore personnel to enter any such spaces, conduct a pre-operational risk assessment prior to commencement of cargo operations or arrival of shore personnel. When hazards are identified a joint risk assessment with terminal representatives responsible for operations on board the ship, or with other appropriate shore personnel, should be conducted; and

.2 identify and agree to the precautions required during entry and enclosed space entry permit arrangements to be used.

6.3 Entry doors or access hatches leading to enclosed spaces should at all times be secured against entry unless the spaces have been risk assessed, atmospherically tested as required and declared safe for entry. The ship may use a system of seals similar to t hose

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required and declared safe for entry. The ship may use a system of seals similar to t hose recommended in the ISPS Code. Suitable portable signage, which is easily understandable by the vessels crew and also by shore personnel engaged on the ship at this time and in the port, indicating the hazards should be posted on entry doors or access hatches leading to an enclosed space. An example of such signage is contained within appendix 3. These signs should be updated when the space becomes safe for entry or when a safe space becomes unsafe.

6.4 A door or hatch cover which is opened to provide natural ventilation of an enclosed space may, wrongly, be taken to be an indication of a safe atmosphere and therefore it is recommended to station an attendant at the entrance or use a mechanical or physical barrier, such as a locked bar or chain positioned across the opening with an attached warning sign, to

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such as a locked bar or chain positioned across the opening with an attached warning sign, to prevent accidental entry. It is recommended that operations such as these are included in the watch handover activity.

6.5 The master or the responsible person should determine that it is safe to enter an enclosed space by ensuring that:

.1 potential hazards have been identified in the risk assessment and as far as possible isolated or made safe;

.2 the space has been thoroughly ventilated by natural or mechanical means to remove any toxic or flammable gases identified in the hazard identification process gases;

.3 the atmosphere of the space has been tested using certified and calibrated instruments to ascertain that the space contains 20.9% oxygen. If the testing

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instruments to ascertain that the space contains 20.9% oxygen. If the testing indicates that the level of oxygen is less than 20.9%, or indicates the presence of even low levels of flammable or toxic gases, then the space should be treated as one where the atmosphere is known or suspected to be unsafe, as described in section 9;

.4 the space has been declared safe for entry and properly illuminated;

.5 a suitable system of communication between all parties for use during entry has been agreed and tested, and the evacuation signals have been agreed upon;

.6 personnel entering the space are wearing personal gas detection equipment that has been properly calibrated and is capable of monitoring the levels of oxygen, carbon monoxide and any other gases identified in the risk assessment;

.7 an attendant has been nominated and properly instructed;

.8 rescue and resuscitation equipment has been positioned , tested and ready

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.7 an attendant has been nominated and properly instructed;
.8 rescue and resuscitation equipment has been positioned , tested and ready for immediate use at the entrance to the space and a rescue plan detailing the rescue arrangements has been agreed. Where ship or shore personnel are working in more than one cargo space at the same time, the rescue equipment should be positioned at a designated central location. In the event of an emergency in any one space, all personnel working in other spaces should be instructed to stop work immediately and exit the space;
.9 personnel are provided with the appropriate personal protective equipment for the entry and subsequent tasks; and
.10 the required permits have been issued, authorizing entry.
6.6 The precautions in sub -paragraphs .6 to .8 of paragraph 6. 5 above may not be applicable to every situation described in this section. The responsible person authorizing entry
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applicable to every situation described in this section. The responsible person authorizing entry should determine whether personal gas detectors, an attendant and the positioning of rescue equipment at the entrance to the space are necessary.

6.7 Only trained and authorized personnel should be assigned the duties of entering, functioning as attendants or acting as members of rescue teams. Ships' crews with rescue and first aid duties should be drilled periodically as required by SOLAS regulation I II/19.3.6 in rescue and first aid procedures. Training should include as a minimum:

.1 identification of the hazards likely to be faced during entry into enclosed spaces and whilst within the space, in particular the rapidity with which

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spaces and whilst within the space, in particular the rapidity with which oxygen may be depleted in a space by corrosion or biological means especially in higher ambient temperatu res. The pace at which oxygen gets depleted increases exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation) or other characteristics or properties of the contents or structure of the space. While higher ambient temperatures can increase the rate of oxygen depletion, it should be assumed that the rate of depletion will always be very rapid, regardless of the ambient temperature;

.2 an explanation of how the pace at which oxygen is depleted will increase exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed;

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forced ventilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed;
.3 identification and use of the various sources of information on the hazards associated with individual solid bulk and liquid bulk cargoes, and the precautions to be adopted when entering spaces containing such cargoes, or their residues;
.4 awareness of the fact that when a person or persons in an enclosed space shows signs of adverse health effects, that they should always assume that these effects are due to an oxygen -depleted or toxic atmosphere in the space, and that they should not enter it themselves;
.5 recognition of the signs of adverse health effects caused by exposure to hazards during entry;
.6 knowledge and experience in the use of personal protective equipment required for entry;
.7 rescue, first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures;
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required for entry;
.7 rescue, first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures;
.8 knowledge and experience in the use of communication equipment and procedures;
.9 knowledge and experience in the use of instruments for measuring the atmosphere;
.10 knowledge and experience in the use of rescue equipment and procedures;
.11 knowledge of the IMO/WHO/ILO Medical First Aid Guide for Use in Accidents involving Dangerous Goods (MFAG), where appropriate; and
.12 knowledge of use of emergency and the first aid equipment for chemical tankers (IBC Code section 14.3) and for gas carriers (IGC Code section 14.3), where appropriate.

6.8 All equipment used in connection with entry should be in good working condition and inspected and tested prior to use.

6.9 As far as practicable, enclosed space entry should be carried out during hours of daylight or normal working hours of the ship to ensure ready availability of reserve personnel in the event of an emergency.

7 TESTING THE ATMOSPHERE

7.1 The gas detection equipment should be appropriate for the cargo that the ship has carried and is carrying considering information provided in the shipper's declaration, the Safety Data Sheets (SDS), the IMDG Code, the IMSBC Code, International Bulk Chemical Code (IBC Code) and the IGC Code. The gas detection equipment, including devices for testing CO_2, should be capable of operating correctly even in oxygen-depleted atmosphere.

7.2 Appropriate testing of the atmosphere of a space should be carried out with properly calibrated equipment by persons trained in the use of the equipment. The manufacturers'

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calibrated equipment by persons trained in the use of the equipment. The manufacturers' instructions should be strictly followed. Testing of the atmosphere of a space should be carried out before any person enters the space and at regular intervals thereafter until all work is completed. Where appropriate, the testing of the atmosphere of a space should be carried out at as many different levels and areas as is necessar y to account for gas stratification, and obtain a representative sample of the atmosphere in the space. In some cases, it may be difficult to test the atmosphere throughout the enclosed space without entering the space (e.g. the bottom landing of a stairway or complex areas of the structure withi n the space) and this should be taken into account.

7.3 All ships should carry at least two sets of gas detection equipment as required by SOLAS regulation XI -1/7, taking into account the Guidelines to facilitate the selection of

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SOLAS regulation XI -1/7, taking into account the Guidelines to facilitate the selection of portable atmosphere testing instruments for enclosed spaces as required by SOLAS regulation XI-1/7 (MSC.1/Circ.1477). Any ship which may carry a cargo capable of generating hazardous vapours and which requires regular entry into the cargo space for cleaning or inspection should carry two sets of gas detection equipment in addition to those required by SOLAS regulation XI-1/7 for assessing the risk to personnel entering the space. The use of flexible hoses or fixed sampling lines, which reach remote areas within the enclosed space, may allow for safe testing without the need to enter the space.

7.4 After completion of a suitable risk assessment targeted to the space to be entered steady readings of all of the following should be obtained:

.1 20.9% oxygen by volume;

.2 the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm);

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.1 20.9% oxygen by volume;
.2 the level of carbon dioxide has been checked and is less than 0.5% by volume (5,000 ppm);
Note: National requirements may differ when determining the safe atmosphere range for gases stated above.
.3 less than 1% of lower flammable limit (LFL) on a suitably sensitive combustible gas indicator, where the assessment has determined that there is potential for flammable gases or vapours; and
.4 less than 50% of the occupational exposure limit (OEL) of any toxic vapours and gases.

7.5 If these conditions cannot be met, additional ventilation should be applied to the space and retesting should be conducted after a suitable interval. Entry should be allowed only after all the above conditions are met. When the atmosphere remains, or is suspected to be unsafe, then the guidance contained within section 9 should apply.

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7.6 Any gas testing should be carried out with ventilation to the enclosed space stopped, and after conditions have stabilized, in order to obtain accurate readings.
7.7 Where the assessment has determined that there is potential for the presence of toxic gases and vapours, appropriate testing should be carried out, using fixed or portable gas or vapour detection and/or monitoring equipment. The readings obtained by this equipment should be below the occupational exposure limits for the toxic gases or vapours given in accepted national or international standards, in accordance with paragraph 7.4. It should be noted that testing for flammability or oxygen content does no t provide a suitable means of measuring for toxicity, nor vice versa.

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noted that testing for flammability or oxygen content does no t provide a suitable means of measuring for toxicity, nor vice versa.

7.8 It should be emphasized that the internal structure of the space, cargo, cargo residues and tank coatings may also present situations where oxygen -deficient areas may exist, and should always be suspected, even when an enclosed space has been satisfactori ly tested as being suitable for entry. This is particularly the case for spaces where the path of the supply and outlet ventilation is obstructed by structural members or cargo.

8 PRECAUTIONS DURING ENTRY

8.1 The atmosphere should be tested frequently whilst the space is occupied, and persons should be instructed to leave the space should there be any sign of deterioration in the conditions.

8.2 Persons entering enclosed spaces should be provided with calibrated and tested personal gas detection instrument or instruments that monitor the levels of oxygen, carbon

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personal gas detection instrument or instruments that monitor the levels of oxygen, carbon dioxide, flammable gases or vapours, toxic gases (including carbon monoxide), and any other gases identified in the risk assessment.

8.3 Ventilation should be maintained while the space is occupied. After a break and before re-entry, the atmosphere must be retested, the results verified as being acceptable then recorded. In the event of failure of the ventilation system, any persons in the space should leave immediately.

8.4 Particular care should be taken when working on piping and valves within the space which may contain hazardous gases, vapours or liquids. If conditions change during the work, increased frequency of testing of the atmosphere should be performed. Examples of changing conditions are as follows increase or decrease in ambient temperature, the need to use oxygen fuel torches or other welding equipment, the need to use a mobile plant, other activities

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oxygen fuel torches or other welding equipment, the need to use a mobile plant, other activities in the space that may involve development of vap our suc h as cleaning debris, collecting sediments, painting within the enclosed spaces, changes in the ship's trim and list while working within the enclosed space. Conditions can also deteriorate during work breaks because of changes to the ventilation, the dist urbance of bilge contents, sludge or stagnant water, or the leakage of external contaminants into the space. Depending on the result of the testing, a decision should be made as to whether it is safe to continue working.

8.5 In the event of an emergency, under no circumstances should the attendant enter the space before help has arrived and the situation has been evaluated to ensure the safety of those entering the space to undertake rescue operations. Only properly trained and equipped personnel should perform rescue operations in enclosed spaces.

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personnel should perform rescue operations in enclosed spaces.

9 ADDITIONAL PRECAUTIONS FOR ENTRY INTO A SPACE WHERE THE ATMOSPHERE IS KNOWN OR SUSPECTED TO BE UNSAFE

9.1 In preparing to enter an enclosed space, every effort should be made to ensure that it is safe to do so in order to undertake the activity required within the space. Entry into enclosed spaces where the atmosphere is known or suspected to be unsafe requires very careful consideration, including an assessment of the hazards, residual risks and mitigations that need to be undertaken. Spaces that have not been tested should be considered unsafe for persons to enter.

9.2 When considering the planned work activity within an enclosed space where the

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9.2 When considering the planned work activity within an enclosed space where the atmosphere is known or suspected to be unsafe, suitably designed and constructed breathing apparatus of positive pressure type should always be worn, and only personnel trained i n its use should be allowed to enter the space. Air -purifying respirators, dust masks and canister face masks do not provide a supply of clean air from a source independent of the atmosphere within the space and should not be used. Emergency Escape Breathing Devices (EEBDs) are not suitable to use for entry into enclosed spaces.
9.3 Notwithstanding paragraph 6.5.6, persons entering enclosed spaces that may contain a suspected atmospheric hazard should be provided with calibrated and tested personal gas detector suitable for the gas or gases assessed as likely to occur in the space.
9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be
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9.4 Rescue harnesses should be worn and unless impractical, lifelines should also be used. A means to facilitate evacuation from the enclosed space should be available and ready for use, as per emergency response plan.
9.5 Appropriate protective clothing should be worn, particularly where there is any risk of toxic substances or chemicals coming into contact with the skin of those entering the space.
9.6 The advice in section 11 concerning actions to be taken in an emergency is particularly relevant in this context.
10 HAZARDS RELATED TO SPECIFIC TYPES OF SHIPS OR CARGO
10.1 Dangerous goods in packaged form
10.1.1 The atmosphere of any space containing dangerous goods may put at risk the health or life of any person making an entry. Dangers may include flammable, toxic, corrosive or asphyxiant gases or vapours, residues on packages and spilled material. The same haz ards may be present in spaces adjacent or connected to the cargo spaces. Information on the
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may be present in spaces adjacent or connected to the cargo spaces. Information on the hazards of specific substances is contained in the IMDG Code, the SDS and the Shippers Declaration. If there is evidence or suspicion that leakage of dangerous sub stances has occurred, the precautions specified in these recommendations should be followed.

10.1.2 Personnel required to deal with spillages or to remove defective or damaged packages should be appropriately trained and wear suitable breathing apparatus and protective clothing commensurate with the task.

10.2 Liquid bulk

The industry has produced extensive advice to managers, operators and crews of ships engaged in the bulk carriage of oil, chemicals and liquefied gases, in the form of specialist safety guides. Information contained in the guides on enclosed space entry am plifies these recommendations. In particular, for chemical tankers as defined in SOLAS regulation VII/8, the

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recommendations. In particular, for chemical tankers as defined in SOLAS regulation VII/8, the diversity of bulk liquid chemicals carried in some cargo tanks and the limitations in vapour detection technologies may pose complications that require specific and targeted mitigation.

10.2.1 Safety information
Safety information2 for the correct handling and carriage of liquid bulk and gas cargoes is set out in the SDS or other cargo information provided by the shipper to the master. Such information should be made available to all ship and shore personnel involved in the handling of the cargo, cargo slops and tank cleaning.
10.2.2 Use of nitrogen as an inert gas3
Nitrogen is a colourless and odourless gas that, when used as an inert gas, causes oxygen
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10.2.2 Use of nitrogen as an inert gas3

Nitrogen is a colourless and odourless gas that, when used as an inert gas, causes oxygen deficiency in enclosed spaces and at exhaust openings on deck during purging of tanks and void spaces and use in cargo holds. It should be noted that one deep breath of 100% nitrogen gas will prove fatal.

10.3 Solid bulk

On ships carrying solid bulk cargoes, dangerous atmospheres may develop within cargo spaces, connected and adjacent spaces. The hazards may include flammability, toxicity, oxygen depletion, carbon dioxide and/or carbon monoxide generation, or self -heating, as identified in the shipper's declaration and/or in the individual schedules in appendix 1 of the IMSBC Code. Solid bulk cargoes listed in the IMSBC Code should be carried in accordance with the provisions of the Code, including precautions, atmosphere t esting, ventilation and other requirements specified. Solid bulk cargoes that are not listed in the IMSBC Code should

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other requirements specified. Solid bulk cargoes that are not listed in the IMSBC Code should be carried in accordance with section 1.3 of the Code, including the conditions for carriage and handling as determined by the relevant co mpetent authorities. Grain cargoes and timber cargoes other than those listed in the IMSBC Code 4 may also cause oxygen depletion and toxic gas emissions, primarily carbon dioxide, in cargo holds and connected and adjacent spaces.

10.3.1 Enclosed hold access trunks

10.3.1.1 In certain designs of hold access, a stair arrangement, sometimes referred to as the "Australian Ladder", substitutes for a vertical ladder. On some ships these are contained within enclosed protective structures that are open only at the top and bottom o f the hold. When a cargo that presents the risk of a hazardous atmosphere is loaded, this "connected" space will quickly adopt atmospheric attributes similar to that of the source space. The enclosed

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quickly adopt atmospheric attributes similar to that of the source space. The enclosed Australian ladder access should not be used unti l it has been fully ventilated, tested and confirmed to be safe. It should be noted that such spaces are difficult to ventilate, unless cargo blocking the bottom of the ladder is removed.

10.3.1.2 When hatches are opened to ventilate the above cargo space at discharge, the hazardous atmosphere often gets trapped in the access trunk.

2 16.2.3.1 of the IBC Code and 18.3.1 of the IGC Code.

3 Refer to the Guidelines on tank entry for tankers using nitrogen as an inerting medium (MSC.1/Circ.1401).

4 Refer to the International Code for the Safe Carriage of Grain in Bulk (resolution MSC.23(59) (amended by resolution MSC.552(108)) and the Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2011 (resolution A.1048(27)).

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